



BITS Pilani
Pilani | Dubai | Goa | Hyderabad

Lab manual

CSIS Dept, WILP

Link for the lab through eLearn portal

The screenshot shows a web browser window with the URL elearn.bits-pilani.ac.in. The page header includes the BITS Pilani logo and the text "WORK INTEGRATED LEARNING PROGRAMMES". A user profile for "Santosh Bangaru" is visible in the top right. The main content area features a background image of a woman studying in a library, with the text "eLearn Explore Courses & Live Lectures" and "Get instant access to course details and live lectures with expert instructors." A blue button labeled "Access Courses & Live Lectures >" is highlighted with a red arrow pointing to it, and a red box with the text "Click on this button" is overlaid on the arrow. A "Steps Recorder" window is open in the top right corner, indicating that the tool is being phased out. The Windows taskbar at the bottom shows the date as 09-10-2025 and the time as 11:09.

Welcome - eLearn Integrated P x +

elearn.bits-pilani.ac.in

WORK INTEGRATED LEARNING PROGRAMMES

Santosh Bangaru

eLearn
Explore Courses & Live Lectures
Get instant access to course details and live lectures with expert instructors.

Access Courses & Live Lectures >

Click on this button

<https://elearn.bits-pilani.ac.in/academics/>

27°C Mostly sunny

Search

ENG US

11:09 09-10-2025

Link for the lab through eLearn portal

The screenshot displays the 'My Academics' page on the eLearn portal. The page features a header with the BITS Pilani logo and 'WORK INTEGRATED LEARNING PROGRAMMES'. Below the header, there are navigation tabs: 'Home', 'My Academics' (which is highlighted), and 'WILP Policies'. The main content area has a large banner with the text 'My Academics' and a description: 'The My Academics page is your central hub for all learning activities, featuring easy access to My Courses, eLibrary, My Virtual Labs, and Viva/Project.' Below the banner, there are three buttons: 'My Virtual Labs' (which is highlighted with a red box and a red arrow pointing to it), 'Viva / Project', and 'Academics Feedback >'. The browser's address bar shows the URL 'elearn.bits-pilani.ac.in/academics/'. A 'Steps Recorder' window is visible in the top right corner, indicating that the page is being recorded. The Windows taskbar at the bottom shows the date as 09-10-2023 and the time as 11:09.

My Academics

The My Academics page is your central hub for all learning activities, featuring easy access to My Courses, eLibrary, My Virtual Labs, and Viva/Project.

My Virtual Labs

Viva / Project

Academics Feedback >

Link for the lab through eLearn portal

My Academics - eLearn Integr...

elearn.bits-pilani.ac.in/academics/#myVirtualLabs

Home My Academics WILP Policies

My Courses | eLibrary | My Virtual Labs | Viva / Project

Academics Feedback >

My Virtual Labs

My Virtual Labs offers students access to a wide range of virtual experiments across various courses, providing hands-on learning opportunities in a simulated environment to enhance their academic experience.

CSIS - Connect to Remote Lab View >

CSIS - Connect to Virtual Lab View >

CSIS Lab - User Manual View >

Click on this button

Steps Recorder - Recording Now

This tool is being phased out. Click here to explore alternatives.

Pause Record Stop and Review Settings

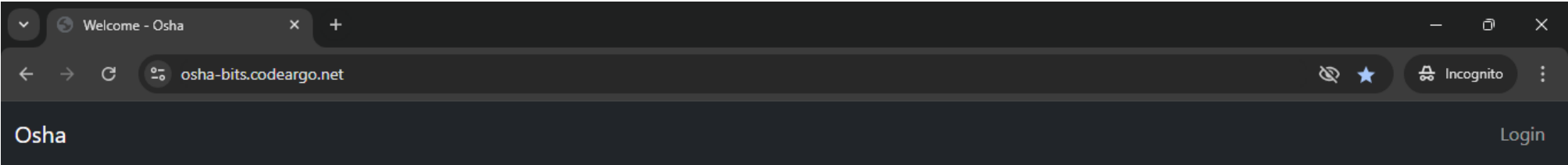
27°C Mostly sunny

Search

ENG US

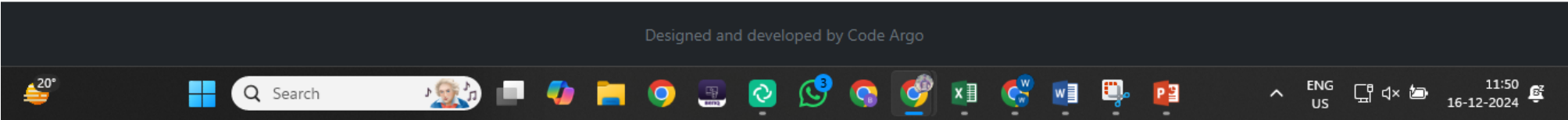
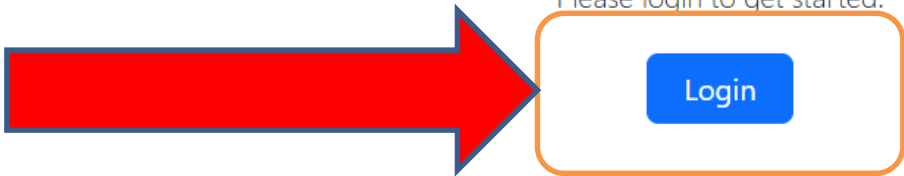
11:09 09-10-2025

Osha Portal Dashboard landing page. Click on Login button to access your lab.



Welcome to Lab Access Portal

Please login to get started.



Enter credentials(Same as eLearn Credentials)

My Academics - eLearn Integrated Portal Login - eLearn Integrated Portal

idp.bits-pilani.ac.in/idp/Authn/UserPassword

WORK INTEGRATED LEARNING PROGRAMMES

BITS Email Address

santosh.bangaru@hyderabad.bits-pilani.ac.in

Password

Remember Me [Forgot password?](#)

Log >

[Click here for login instructions](#)

An institution deemed to be a University estd. vide Sec.3 of the UGC Act, 1956 under notification #F.12-23/63.U-2 of June 18, 1964

innovate achieve lead

Steps Recorder - Recording Now

This tool is being phased out. [Click here](#) to explore alternatives.

Pause Record Stop and Review Settings

27°C Mostly sunny

Search

ENG US

11:09 09-10-2025

You are now successfully launched lab portal.

Lab Access Booking - Osha

osha-bits.codeargo.net/booking/

School

OshaWILP CSIS Lab

Course / Experiment:

Date:

Slot:

Book

--- Select Course / Experiment ---

--- Select Date ---

--- Select Slot ---

Course	Machine	Time Used	Time Remaining	State	Control
Big Data Systems (Cluster Lab) (Master)	i-0e4aebdc0646e60e9	12m 59s	21h 47m	saved	Start
Big Data Systems (Cluster Lab) (Node-1)	i-0ec0e002c81695ddc	10m 12s	21h 49m	saved	Start
Big Data Systems (Cluster Lab) (Node-2)	i-0e73fc5ce538079ce	11m 1s	21h 48m	saved	Start

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29°C
Haze

Search

ENG
US

12:12
09-10-2025

Select the type of lab that you are accessing and click on the Icon



Cluster Lab



Console Lab



Virtual Lab



Remote Lab



Cluster Lab

ज्ञानं परमं बलम्

Select the course and click on “Launch” button to launch the Cluster machines

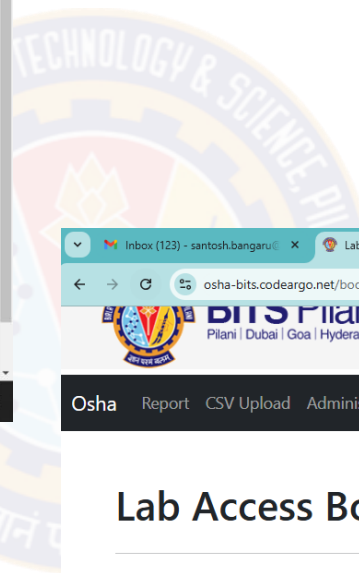
Lab Access Booking

Course / Experiment: --- Select Course / Experiment --- Date: --- Select Date --- Slot: --- Select Slot --- Book

--- Select Course / Experiment ---

- APACHE HADOOP CLUSTER (OPENSTACK)
- APACHE KAFKA (OPENSTACK)
- APACHE SPARK (OPENSTACK)
- APACHE STORM (OPENSTACK)
- COCKROACHDB CLUSTER (OPENSTACK)
- MONGODB CLUSTER (OPENSTACK)
- MPI CLUSTER (OPENSTACK)
- NFS CLIENT-SERVER (OPENSTACK)
- POSTGRESCL CLUSTER (OPENSTACK)
- Testing tools-Appsv1
- DOCKER (OPENSTACK)

Machine	Start Time	End Time	Slot Counted
APACHE HADOOP	Nov. 18, 2024, 8 a.m.	Nov. 18, 2024, 10 a.m.	✓
APACHE HADOOP	Nov. 21, 2024, 10 a.m.	Nov. 21, 2024, noon	✓
APACHE HADOOP	Nov. 25, 2024, 10 a.m.	Nov. 25, 2024, noon	✓
APACHE HADOOP	Nov. 27, 2024, 10 a.m.	Nov. 27, 2024, noon	✓
DOCKER	Nov. 27, 2024, 2 p.m.	Nov. 27, 2024, 3 p.m.	✓
APACHE HADOOP CLUSTER (OPENSTACK)	Dec. 6, 2024, 8 p.m.	Dec. 6, 2024, 10 p.m.	✓
APACHE HADOOP CLUSTER (OPENSTACK)	Dec. 14, 2024, noon	Dec. 14, 2024, 2 p.m.	✓



Lab Access Booking

Course / Experiment: Applied Machine Learning_SESAP Date: --- N/A

Click on “Launch”

Launch

Past Bookings in the last 30 days

Course	Machine	Start Time	End Time	Slot Counted
Big Data Analytics (S2-24_BAZG525) - Apache Framework	APACHE FRAMEWORK - 1	Feb. 6, 2025, 10 a.m.	Feb. 6, 2025, noon	✓
Big Data Analytics (S2-24_BAZG525) - Apache Framework	APACHE FRAMEWORK - 1	Feb. 6, 2025, 4 p.m.	Feb. 6, 2025, 6 p.m.	✓

After clicking launch, you can now see the cluster machines as shown in the following screenshot. Click on “Start” button to start the machines

Lab Access Booking - Osha

osha-bits.codeargo.net/booking/

Osha WILP CSIS Lab

Course / Experiment: Date: Slot:

--- Select Course / Experiment --- --- Select Date --- --- Select Slot --- Book

Course	Machine	Time Used	Time Remaining	State	Control
Big Data Systems (Cluster Lab) (Master)	i-0e4aebdc0646e60e9	12m 59s	21h 4		Start
Big Data Systems (Cluster Lab) (Node-1)	i-0ec0e002c81695ddc	10m 12s	21h 49m	saved	Start
Big Data Systems (Cluster Lab) (Node-2)	i-0e73fc5ce538079ce	11m 1s	21h 48		Start

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29°C Haze Search ENG US 12:12 09-10-2025

After clicking on “Start” button, Kindly wait for the machines to be launched for accessing.

The screenshot shows the Osha Lab Access Booking interface. At the top, there are dropdown menus for 'Course / Experiment', 'Date', and 'Slot', followed by a 'Book' button. Below this, a light blue notification bar states: 'A fresh AWS instance was launched for you. Please wait for a few minutes before connecting.' Below the notification is a table with the following data:

Course	Machine	Time Used	Time Remaining	State	Control
Big Data Systems (Cluster Lab) (Master)	i-0a337bdb5ad6de95d	14m 11s	21h 45m	pending	Please wait ...
Big Data Systems (Cluster Lab) (Node-1)	i-0dcc378092ff7e5b5	11m 17s	21h 48m	pending	Please wait ...
Big Data Systems (Cluster Lab) (Node-2)	i-0e4d0ebde97ae3c50	12m 5s	21h 47m	pending	Please wait ...

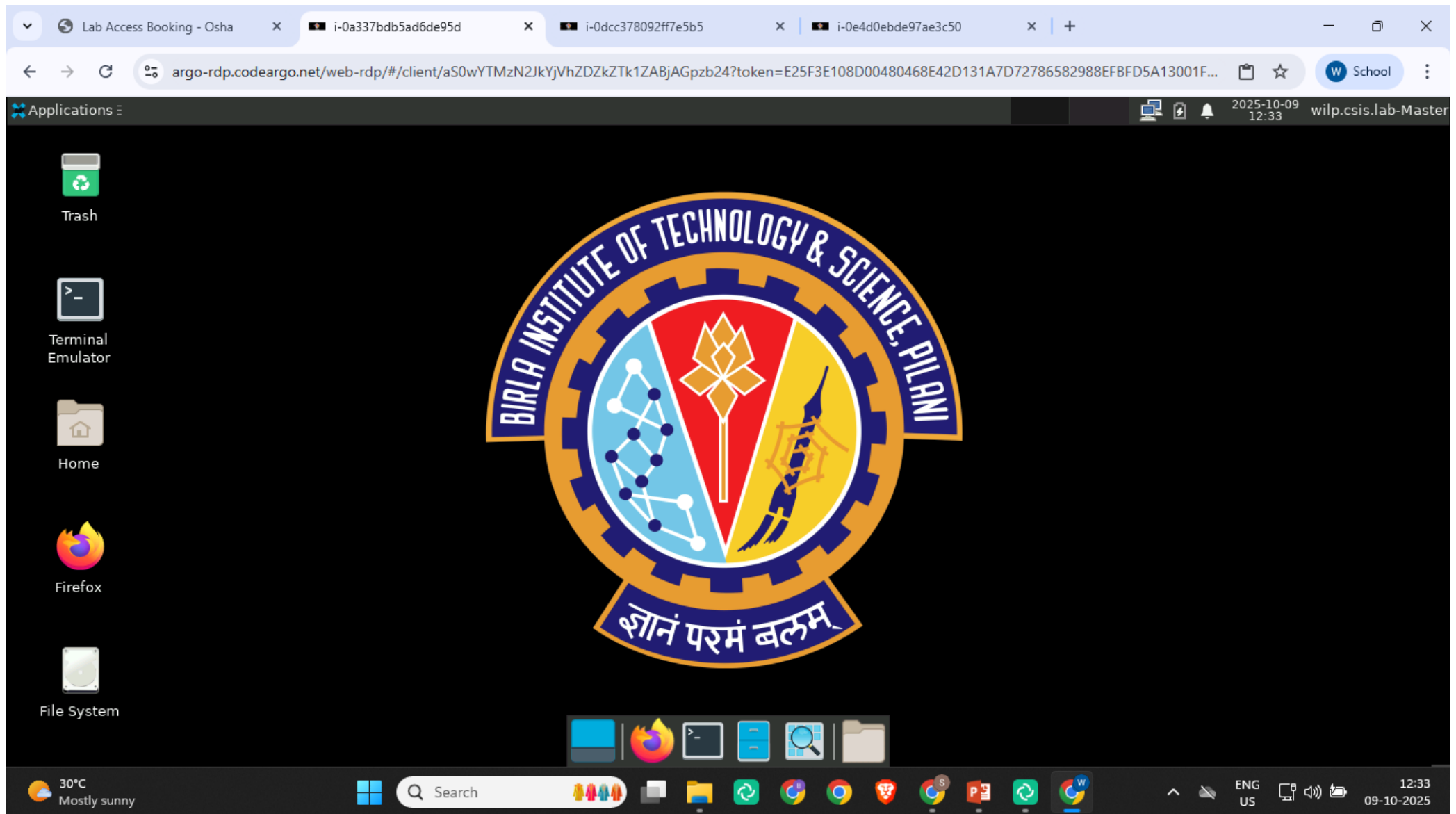
Below the table, there is a section for 'Upcoming Bookings' with columns for Course, Machine, Start Time, End Time, Connect, and Cancel.

This screenshot shows the same Osha Lab Access Booking interface, but the machines are now in a 'started' state. The notification bar remains. The table data is as follows:

Course	Machine	Time Used	Time Remaining	State	Control
Big Data Systems (Cluster Lab) (Master)	i-0a337bdb5ad6de95d	14m 11s	21h 45m	started	Connect Stop
Big Data Systems (Cluster Lab) (Node-1)	i-0dcc378092ff7e5b5	11m 17s	21h 48m	started	Connect Stop
Big Data Systems (Cluster Lab) (Node-2)	i-0e4d0ebde97ae3c50	12m 5s	21h 47m	started	Connect Stop

A red arrow points from the text 'Click on “Connect” to access the machines' to the 'Connect' buttons in the 'Control' column. Below the table is the 'Upcoming Bookings' section.

You are now successfully launched your lab machine



Kindly Stop all the lab machines once your lab work is completed.

Lab Access Booking - Osha

osha-bits.codeargo.net/booking/

Osha

WILP CSIS Lab

Course / Experiment:

Date:

Slot:

Book

A fresh AWS instance was launched for you. Please wait for a few minutes before connecting.

Course	Machine	Time Used	Time Remaining	State	Control
Big Data Systems (Cluster Lab) (Master)	i-0a...				Connect
Big Data Systems (Cluster Lab) (Node-1)	i-0d...				Connect
Big Data Systems (Cluster Lab) (Node-2)	i-0e4d0ebde97ae3c50	12m 36s	21h 47m	started	Connect

Upcoming Bookings

Course	Machine	Start Time	End Time	Connect	Cancel
--------	---------	------------	----------	---------	--------

30°C Mostly sunny

Search

ENG US

09-10-2025

12:31

Click on "Stop"

Lab Access Booking - Osha

osha-bits.codeargo.net/booking/

Osha

WILP CSIS Lab

Course / Experiment:

Date:

Slot:

Book

Course	Machine	Time Used	Time Remaining	State	Control
Big Data Systems (Cluster Lab) (Master)	i-0a337bdb5ad6de95d	20m	21h	stopping	Please wait ...
Big Data Systems (Cluster Lab) (Node-1)	i-0d...			stopping	Please wait ...
Big Data Systems (Cluster Lab) (Node-2)	i-0e4d0ebde97ae3c50	12m 36s	21h 47m	stopping	Please wait ...

Upcoming Bookings

Course	Machine	Start Time	End Time	Connect	Cancel
Video Analytics (Remote Lab)	GPU CLUSTER8	Oct. 9, 2025, noon	Oct. 9, 2025, 2 p.m.	Connect	Cancel

30°C Mostly sunny

Search

ENG US

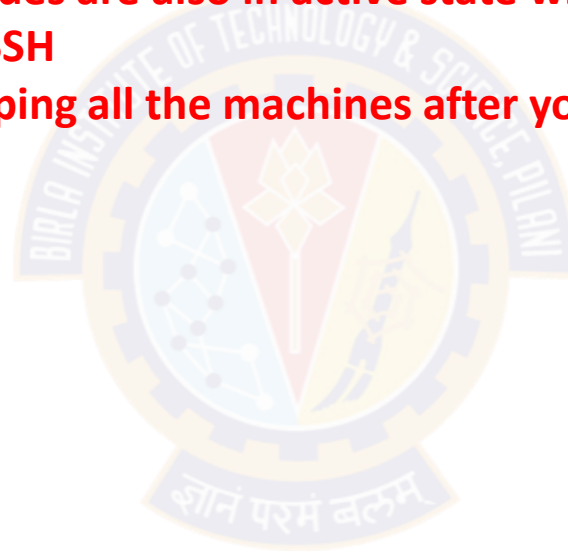
09-10-2025

12:37

State is now "Stopping"

Important instructions

- SSH access is enabled in all the cluster machines. You may use the worker nodes by directly connecting to them or you can SSH into them through Master Node. Make sure the worker nodes are also in active state while you are trying to accessing them through the SSH
- Please ensure that you are stopping all the machines after your work is completed



Console Lab

ज्ञानं परमं बलम्

Select the course and click on “Launch” to launch the AWS Console

Lab Access Booking

Course / Experiment: --- Select Course / Experiment --- Date: --- Select Date --- Slot: --- Select Slot --- [Book](#)

- Select Course / Experiment ---
- APACHE HADOOP CLUSTER (OPENSTACK)
- APACHE KAFKA (OPENSTACK)
- APACHE SPARK (OPENSTACK)
- APACHE STORM (OPENSTACK)
- COCKROACHDB CLUSTER (OPENSTACK)
- MONGODB CLUSTER (OPENSTACK)
- MPI CLUSTER (OPENSTACK)
- NFS CLIENT-SERVER (OPENSTACK)
- POSTGRESCL CLUSTER (OPENSTACK)
- Testing tools-Appsv1
- DOCKER (OPENSTACK)

	Machine	Start Time	End Time	Slot Counted
	APACHE HADOOP	Nov. 18, 2024, 8 a.m.	Nov. 18, 2024, 10 a.m.	✓
	APACHE HADOOP	Nov. 21, 2024, 10 a.m.	Nov. 21, 2024, noon	✓
	APACHE HADOOP	Nov. 25, 2024, 10 a.m.	Nov. 25, 2024, noon	✓
	APACHE HADOOP	Nov. 27, 2024, 10 a.m.	Nov. 27, 2024, noon	✓
	DOCKER	Nov. 27, 2024, 2 p.m.	Nov. 27, 2024, 3 p.m.	✓
	APACHE HADOOP	Dec. 6, 2024, 8 p.m.	Dec. 6, 2024, 10 p.m.	✓
	APACHE HADOOP	Dec. 14, 2024, noon	Dec. 14, 2024, 2 p.m.	✓

Lab Access Booking

Course / Experiment: Applied Machine Learning_SESAP Date: --- N/A [Launch](#)

Click on “Launch”

Past Bookings in the last 30 days

Course	Machine	Start Time	End Time	Slot Counted
Big Data Analytics (S2-24_BAZG525) - Apache Framework	APACHE FRAMEWORK - 1	Feb. 6, 2025, 10 a.m.	Feb. 6, 2025, noon	✓
Big Data Analytics (S2-24_BAZG525) - Apache Framework	APACHE FRAMEWORK - 1	Feb. 6, 2025, 4 p.m.	Feb. 6, 2025, 6 p.m.	✓

Upon launch, Click on “Launch” to access the AWS Console

o sha-bits.codeargo.net/booking/

Osha Report CSV Upload Administration Santosh Bangaru

Course / Experiment: Date: Slot:

--- Select Course / Experiment --- --- Select Date --- --- Select Slot --- Book

Course	Credits ⓘ	State	Control
S1-25-NSPCC-CC	\$0.00 / \$10.00 (0.00%)		Launch
Cloud Computing Testing (console lab)	\$0.00 / \$2.00 (0.00%)	success	Connect

Click on Launch

Course	Machine	Time Used	Time Remaining	State	Control
S2-24-NSP2-AIML Robo - Ubuntu	i-072add1e62a260b1d	35m 38s	21h 24m	saved	Start

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30°C Partly sunny Search ENG US 15:31 07-10-2025

On clicking the Connect button, you will be landed on to the AWS Console page

The screenshot shows the AWS Console Home page. The browser address bar displays `ap-south-1.console.aws.amazon.com/console/home?region=ap-south-1`. The top navigation bar includes the AWS logo, a search bar, and the current region `Asia Pacific (Mumbai)`. The account ID `6304-9302-9178` and the user `santosh.bangaru/santosh.bangaru-cloud-computing...` are visible in the top right.

The main content area is titled **Console Home** and features two primary sections:

- Recently visited**: A section with a placeholder icon and the text "No recently visited services". It suggests exploring common services like [EC2](#), [S3](#), [Aurora and RDS](#), and [Lambda](#). A [View all services](#) link is at the bottom.
- Applications (0)**: A section for managing applications. It shows the current region as `ap-south-1 (Current Region)` and a search bar labeled `Find applications`. Below this is a table header with columns: `Name`, `Description`, `Region`, and `Originati...`. The table is currently empty, displaying the message "No applications. Get started by creating an application." with a [Create application](#) button. A [Go to myApplications](#) link is at the bottom.

The bottom of the page includes a `CloudShell` button, a `Feedback` link, and a footer with copyright information for Amazon Web Services, Inc. or its affiliates. The system tray at the very bottom shows the date and time as `15:34 07-10-2025` and the language set to `ENG US`.

Signing out from the AWS Console

The screenshot shows the AWS Console interface in the Asia Pacific (Mumbai) region. The user menu is open, displaying account information and a 'Sign out' button. A red arrow points to the 'Sign out' button with the text 'Click here to Signout'.

Account ID: 6304-9302-9178
Account color: Unset
Federated user: santosh.bangaru/santosh.bangaru-cloud-computing-session

Account
Organization
Service Quotas
Billing and Cost Management

Turn on multi-session support
Sign out

Console Home Info

Recently visited Info

No recently visited services

Explore one of these commonly visited AWS services.

EC2 S3 Aurora and RDS Lambda

Applications (0) Info

Region: Asia Pacific (Mumbai)

Select Region
ap-south-1 (Current Region)

Name Description

No applications

View all services

Go to myApplications

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Instructions to follow:

- Please ensure that all services you used are properly terminated once your work is completed.
 - Lab credits will continue to be consumed if services are left running. (Note: Make sure to terminate the EC2 instances instead of merely stopping them, since the attached EBS volumes will continue to incur charges even in stopped state.)
 - The AWS console session remains active for one hour after you log in. To continue, simply click the Launch button again in the OSHA portal.
 - No extra credits shall be provided
 - On 100% usage of given credits, account shall be locked down permanently and you are no longer will be able to access the AWS console. All the services consumed shall be wiped out.
 - Regularly monitor your email for credit usage alerts. You can also check your consumed credits in the OSHA portal under your booking, as shown in the screenshot below.
- (Note: AWS updates credit usage within 24–48 hours of consumption.)

The screenshot displays the OSHA portal interface. At the top, there's a navigation bar with 'Osha', 'Report', 'CSV Upload', and 'Administration'. The user 'Santosh Bangaru' is logged in. Below the navigation bar, there are filters for 'Course / Experiment', 'Date', and 'Slot', each with a dropdown menu, and a 'Book' button. The main content area shows a table with columns: Course, Machine, Credits, State, and Control. A red arrow points to the 'Credits' column, which is highlighted with a red box. The 'Credits' column shows two rows: one with '\$0.00 / \$10.00 (0.00%)' and 'expired' state, and another with '\$0.00 / \$2.00 (0.00%)' and 'success' state. Below this, there's another table with columns: Course, Machine, Time Used, Time Remaining, State, and Control. The first row shows 'S2-24-NSP2-AIML Robo - Ubuntu', 'i-072add1e62a260b1d', '35m 38s', '21h 24m', 'saved' state, and a 'Start' button. The bottom of the screenshot shows a Windows taskbar with the date '07-10-2025' and time '15:45'.

Course	Machine	Credits	State	Control
		\$0.00 / \$10.00 (0.00%)	expired	Launch
		\$0.00 / \$2.00 (0.00%)	success	Connect

Course	Machine	Time Used	Time Remaining	State	Control
S2-24-NSP2-AIML Robo - Ubuntu	i-072add1e62a260b1d	35m 38s	21h 24m	saved	Start

Virtual Lab

ज्ञानं परमं बलम्

Select the course and click on “Book” button to book your lab machine.

Lab Access Booking

Course / Experiment: --- Select Course / Experiment --- Date: --- Select Date --- Slot: --- Select Slot --- Book

--- Select Course / Experiment ---

- APACHE HADOOP CLUSTER (OPENSTACK)
- APACHE KAFKA (OPENSTACK)
- APACHE SPARK (OPENSTACK)
- APACHE STORM (OPENSTACK)
- COCKROACHDB CLUSTER (OPENSTACK)
- MONGODB CLUSTER (OPENSTACK)
- MPI CLUSTER (OPENSTACK)
- NFS CLIENT-SERVER (OPENSTACK)
- POSTGRESCL CLUSTER (OPENSTACK)
- Testing tools-Appsv1
- DOCKER (OPENSTACK)

Machine	Start Time	End Time	Slot Counted
APACHE HADOOP	Nov. 18, 2024, 8 a.m.	Nov. 18, 2024, 10 a.m.	✓
APACHE HADOOP	Nov. 21, 2024, 10 a.m.	Nov. 21, 2024, noon	✓
APACHE HADOOP	Nov. 25, 2024, 10 a.m.	Nov. 25, 2024, noon	✓
APACHE HADOOP	Nov. 27, 2024, 10 a.m.	Nov. 27, 2024, noon	✓
DOCKER	Nov. 27, 2024, 2 p.m.	Nov. 27, 2024, 3 p.m.	✓
APACHE HADOOP CLUSTER (OPENSTACK)	Dec. 6, 2024, 8 p.m.	Dec. 6, 2024, 10 p.m.	✓
APACHE HADOOP CLUSTER (OPENSTACK)	Dec. 14, 2024, noon	Dec. 14, 2024, 2 p.m.	✓

Lab Access Booking

Course / Experiment: Applied Machine Learning_SESAP Date: --- N/A

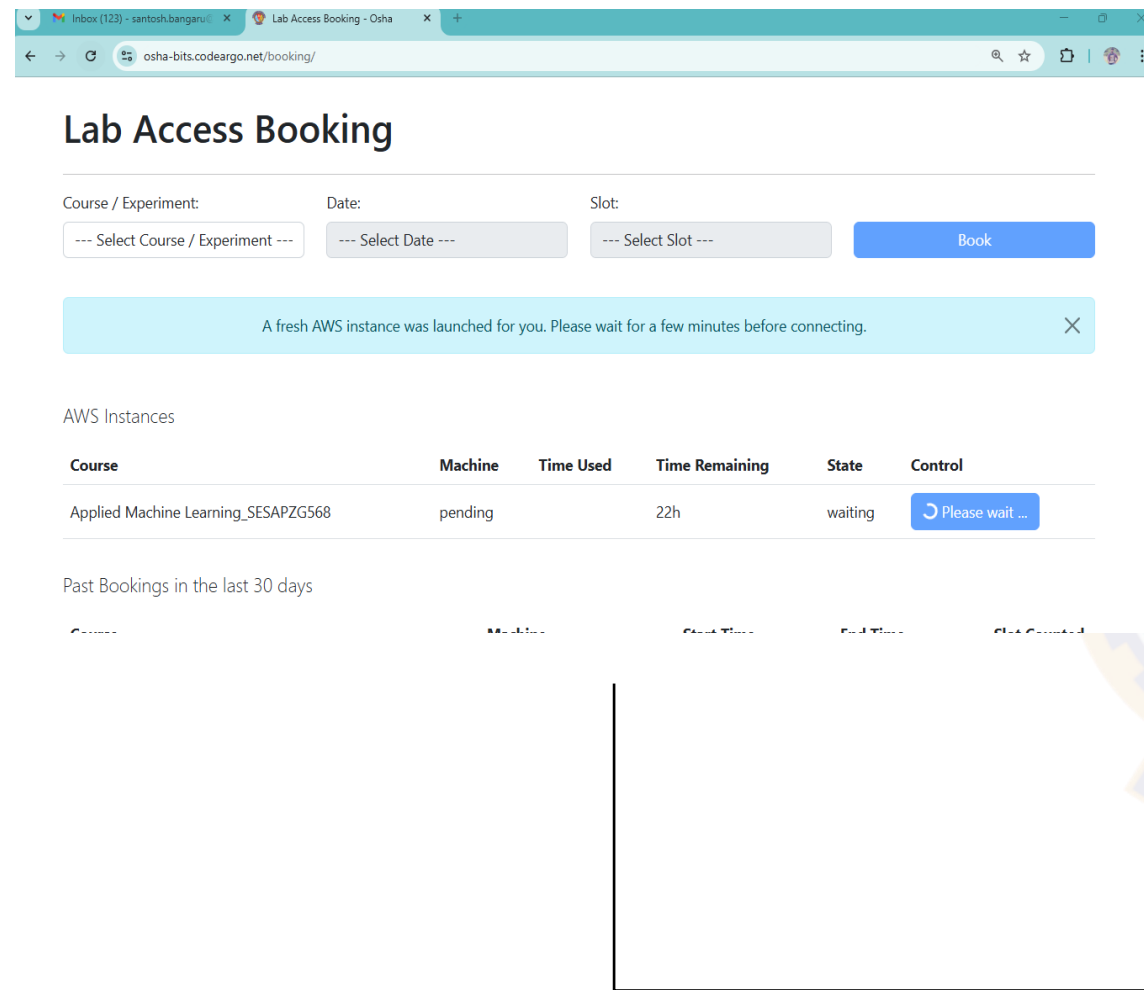
Click on “Launch”

Launch

Past Bookings in the last 30 days

Course	Machine	Start Time	End Time	Slot Counted
Big Data Analytics (S2-24_BAZG525) - Apache Framework	APACHE FRAMEWORK - 1	Feb. 6, 2025, 10 a.m.	Feb. 6, 2025, noon	✓
Big Data Analytics (S2-24_BAZG525) - Apache Framework	APACHE FRAMEWORK - 1	Feb. 6, 2025, 4 p.m.	Feb. 6, 2025, 6 p.m.	✓

Upon launching the machine, Click on “Connect” to access the lab



The screenshot shows the 'Lab Access Booking' page. At the top, there are three dropdown menus for 'Course / Experiment', 'Date', and 'Slot', each with a placeholder '--- Select ... ---'. To the right of these is a blue 'Book' button. Below the dropdowns is a light blue notification box that says 'A fresh AWS instance was launched for you. Please wait for a few minutes before connecting.' with a close button (X). Under the heading 'AWS Instances', there is a table with columns: Course, Machine, Time Used, Time Remaining, State, and Control. The table contains one row for 'Applied Machine Learning_SESAPZG568' with a 'pending' machine, '22h' time remaining, and a 'waiting' state. The 'Control' column for this row has a blue button that says 'Please wait ...'. Below the table, there is a section for 'Past Bookings in the last 30 days' with a table that is partially visible. A black arrow points from the 'Please wait ...' button in the 'AWS Instances' table to the 'Connect' button in the second screenshot.

Lab Access Booking

Course / Experiment: --- Select Course / Experiment --- Date: --- Select Date --- Slot: --- Select Slot --- Book

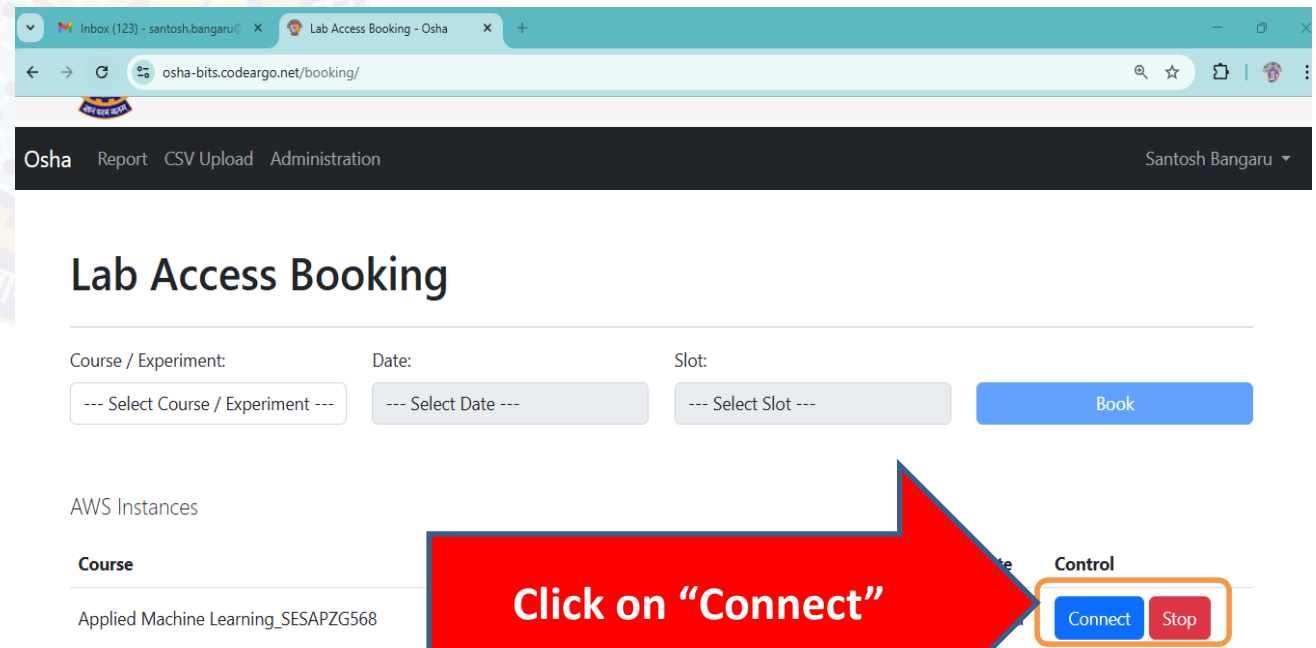
A fresh AWS instance was launched for you. Please wait for a few minutes before connecting.

AWS Instances

Course	Machine	Time Used	Time Remaining	State	Control
Applied Machine Learning_SESAPZG568	pending		22h	waiting	Please wait ...

Past Bookings in the last 30 days

Course	Machine	Start Time	End Time	Slot Counted
--------	---------	------------	----------	--------------



The screenshot shows the 'Lab Access Booking' page. At the top, there are three dropdown menus for 'Course / Experiment', 'Date', and 'Slot', each with a placeholder '--- Select ... ---'. To the right of these is a blue 'Book' button. Below the dropdowns is a light blue notification box that says 'A fresh AWS instance was launched for you. Please wait for a few minutes before connecting.' with a close button (X). Under the heading 'AWS Instances', there is a table with columns: Course, Machine, Time Used, Time Remaining, State, and Control. The table contains one row for 'Applied Machine Learning_SESAPZG568' with a 'pending' machine, '22h' time remaining, and a 'waiting' state. The 'Control' column for this row has two buttons: 'Connect' and 'Stop'. A large red arrow points from the 'Connect' button to the text 'Click on “Connect”'.

Lab Access Booking

Course / Experiment: --- Select Course / Experiment --- Date: --- Select Date --- Slot: --- Select Slot --- Book

AWS Instances

Course	Machine	Time Used	Time Remaining	State	Control
Applied Machine Learning_SESAPZG568	pending		22h	waiting	Connect Stop

Click on “Connect”

On clicking the Username and Password, you can copy the credentials(In case if the machine gets locked,). Click on “Web Console” to access the Machine

Inbox (123) - santosh.bangaru@

Lab Access Booking - Osha

osha-bits.codeargo.net/booking/

Osha

Report

CSV Upload

Administration

Santosh Bangaru

Lab Access Bookin

Course / Experiment:
--- Select Course / Experiment ---

Book

Connect

Please use the following credentials once you've launched the connection:

Username: cloud
Password: cloud

Click here

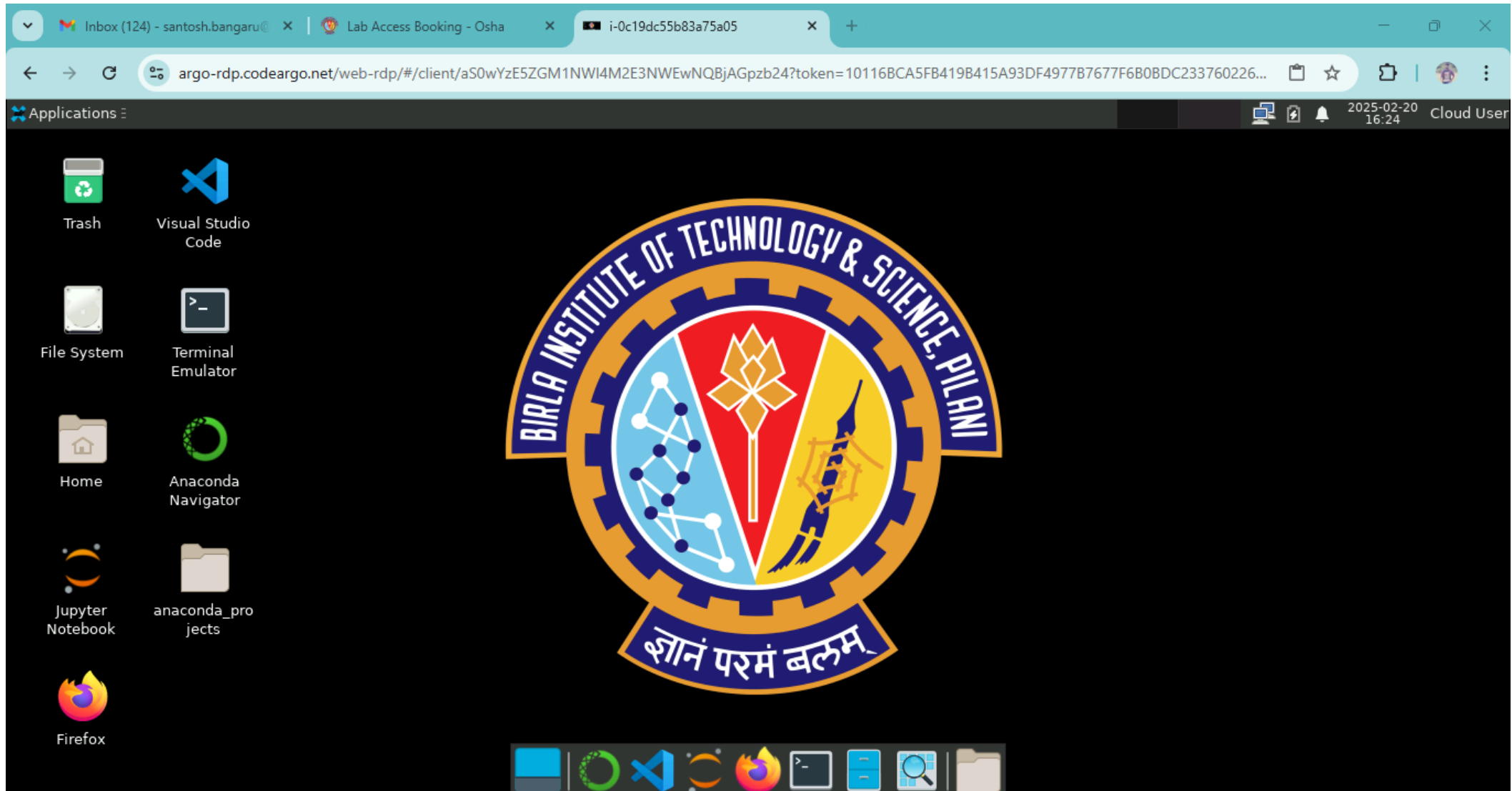
Web Console

AWS Instances

Course	Machine	Time Used	Time Remaining	State	Control
Applied Machine Learning_SESAPZG568	i-0c19dc55b83a75a05	20m 50s	21h 39m	started	Connect Stop

Past Bookings in the last 30 days

You are now successfully logged into the Lab Machine



STOP the machine

osha-bits.codeargo.net/booking/

Lab Access Booking

Course / Experiment: Date: Slot: [Book](#)

AWS Instances

Course	Machine	Time	Control
Applied Machine Learning_SESAPZG568	i-0c19dc55b83a75a05	32	Stop

Past Bookings in the last 30 days

Course	Machine	Start Time	End Time	Slot Counted
Big Data Analytics (S2-24_BAZG525) - Apache Framework	APACHE FRAMEWORK - 1	Feb. 6, 2025, 10 a.m.	Feb. 6, 2025, noon	✓
Big Data Analytics (S2-24_BAZG525) - Apache Framework	APACHE FRAMEWORK - 1	Feb. 6, 2025, 4 p.m.	Feb. 6, 2025, 6 p.m.	✓

Click here to STOP
the machine

Note: Kindly STOP the machine upon completing your work. The lab hours shall be consumed automatically if you forget to STOP the lab. No extra lab hours shall be provided.

Remote Lab

ज्ञानं परमं बलम्

Select the course and click on “Book” button to book your lab machine.

Lab Access Booking - Osha

osha-bits.codeargo.net/booking/

Osha

WILP CSIS Lab

Course / Experiment: --- Select Course / Experiment ---
Date: --- Select Date ---
Slot: --- Select Slot ---

Big Data Systems (Cluster Lab)
Cloud Computing (Console Lab)
Introduction to DevOps (Virtual Lab)
Video Analytics (Remote Lab)
Big Data Systems (Cluster Lab) (Node-1)
Big Data Systems (Cluster Lab) (Node-2)

Machine	Time Used	Time Remaining	State	Control
i-0a337bdb5ad6de95d	22m 47s	21h 37m	saved	Start
i-0dcc378092ff7e5b5	19m 33s	21h 40m	saved	Start
i-0e4d0ebde97ae3c50	20m 21s	21h 39m	saved	Start

Past Bookings in the last 30 days

Course	Machine	Start Time	End Time	Slot Counted
Video Analytics (Remote Lab)	GPU CLUSTER8	Oct. 9, 2025, noon	Oct. 9, 2025, 2 p.m.	✓

23°C Cloudy

Search

11:55 28-10-2025

Select Course, Date and slot

Lab Access Booking - Osha

osha-bits.codeargo.net/booking/

Osha

WILP CSIS Lab

Course / Experiment: Video Analytics (Remote Lab)
Date: --- Select Date ---
Slot: --- Select Slot ---

Click here

Book

Course (Virtual Lab)

Machine	Time Used	Time Remaining	State	Control
i-0a337bdb5ad6de95d	22m 47s	21h 37m	saved	Start
i-0dcc378092ff7e5b5	19m 33s	21h 40m	saved	Start
i-0e4d0ebde97ae3c50	20m 21s	21h 39m	saved	Start

Past Bookings in the last 30 days

Course	Machine	Start Time	End Time	Slot Counted
Video Analytics (Remote Lab)	GPU CLUSTER8	Oct. 9, 2025, noon	Oct. 9, 2025, 2 p.m.	✓

Trending videos

Search

11:58 28-10-2025

You have now booked your slot and you can see it under “Upcoming bookings” section

Lab Access Booking - Osha

osha-bits.codeargo.net/booking/

W School

Course (Virtual Lab)	Machine	Time Used	Time Remaining	State	Control
Big Data Systems (Cluster Lab) (Master)	i-0a337bdb5ad6de95d	22m 47s	21h 37m	saved	<button>Start</button>
Big Data Systems (Cluster Lab) (Node-1)	i-0dcc378092ff7e5b5	19m 33s	21h 40m	saved	<button>Start</button>
Big Data Systems (Cluster Lab) (Node-2)	i-0e4d0ebde97ae3c50	20m 21s	21h 39m	saved	<button>Start</button>

Upcoming Bookings

Course	Machine	Start Time	End Time	Connect	Cancel
Video Analytics (Remote Lab)	GPU CLUSTER9			<button>Connect</button>	<button>Cancel</button>

Past Bookings in the last 30 days

Course	Machine	Start Time	End Time	Slot Counted
Video Analytics (Remote Lab)	GPU CLUSTER8	Oct. 9, 2025, noon	Oct. 9, 2025, 2 p.m.	✓

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23°C Cloudy

Search

W

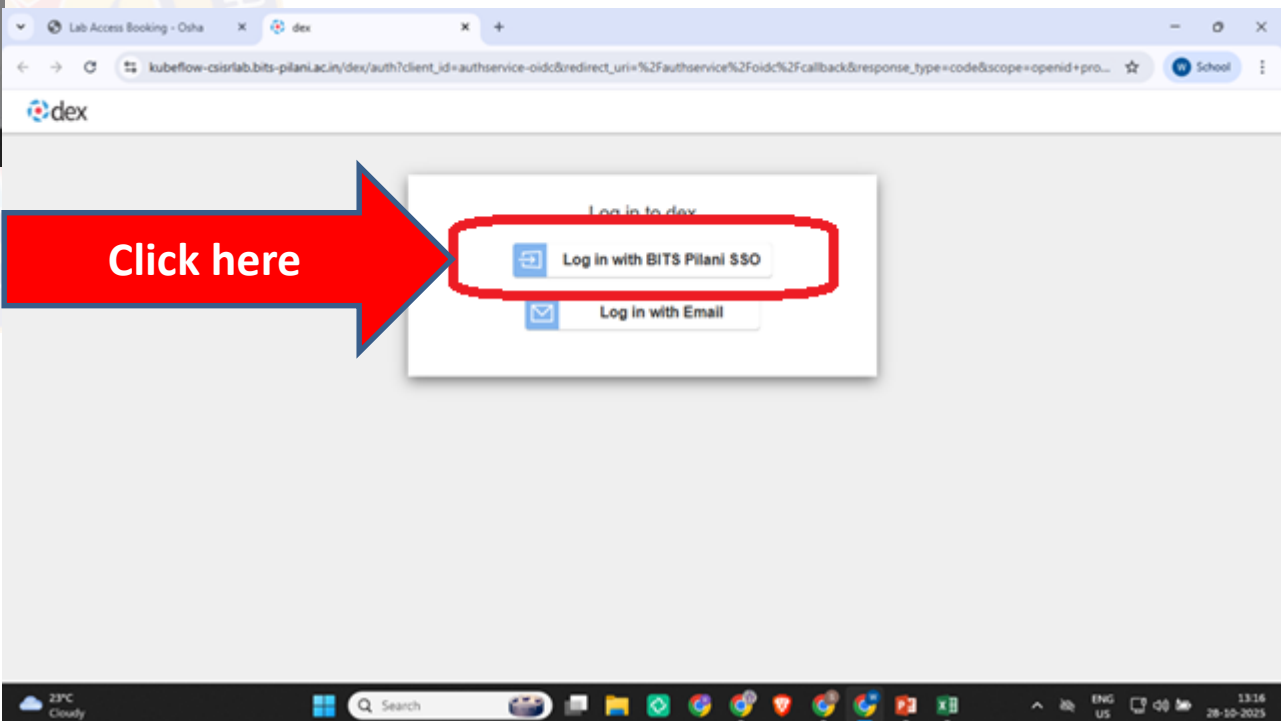
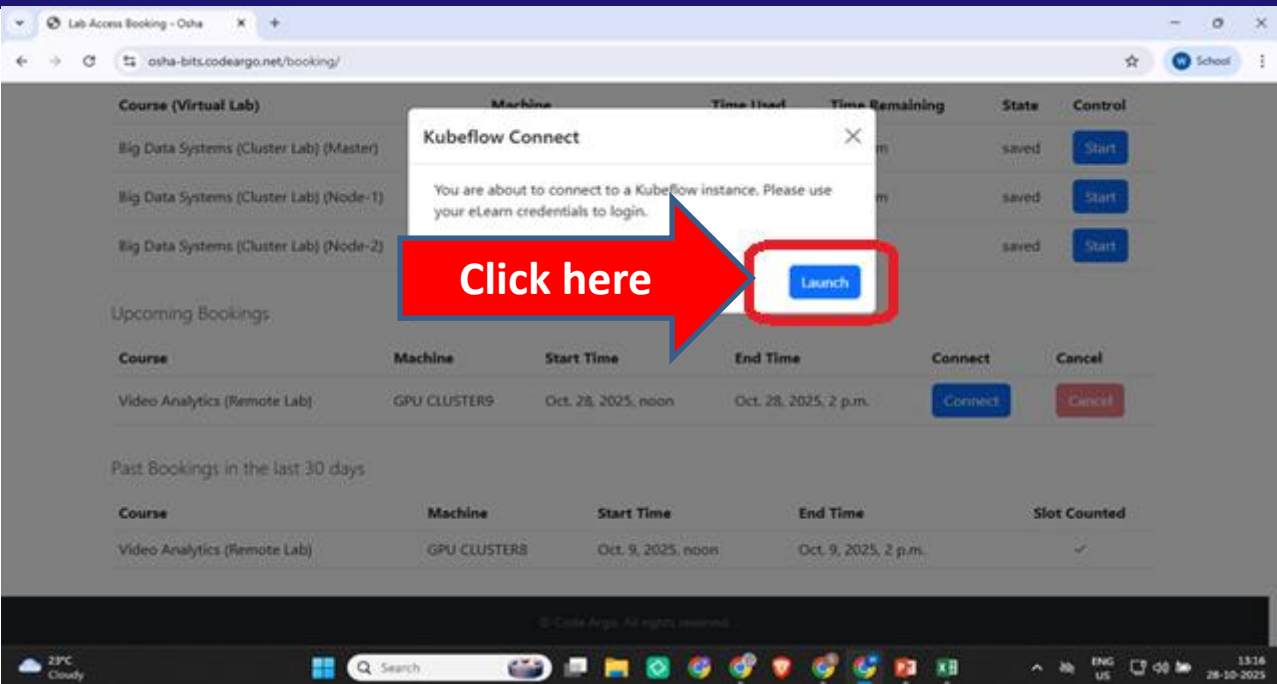
P

X

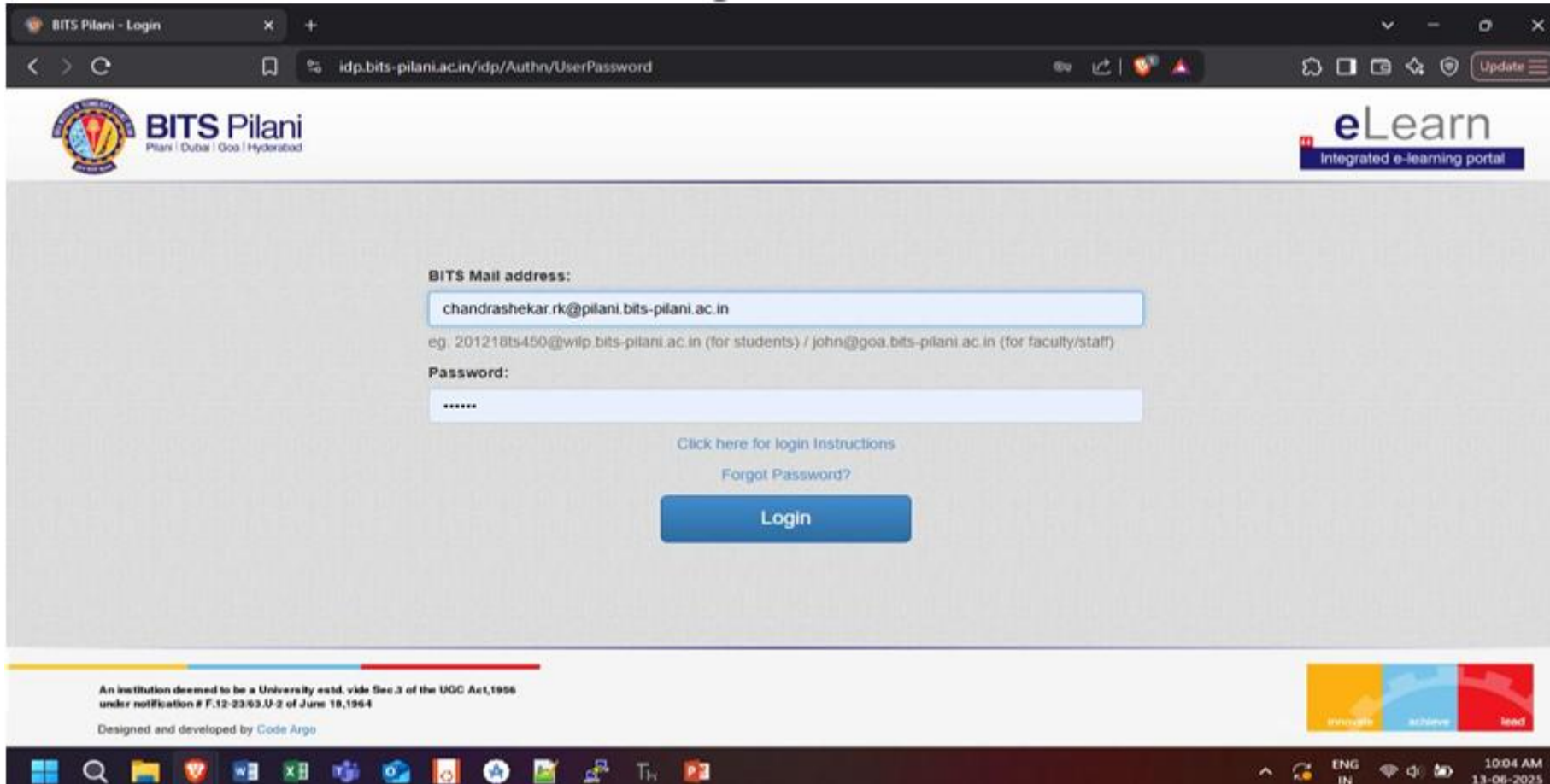
ENG US

13:15 28-10-2025

Click on the “Launch” button and Click on “Login with BITS Pilani SSO”



Enter eLearn portal's user credentials and Login



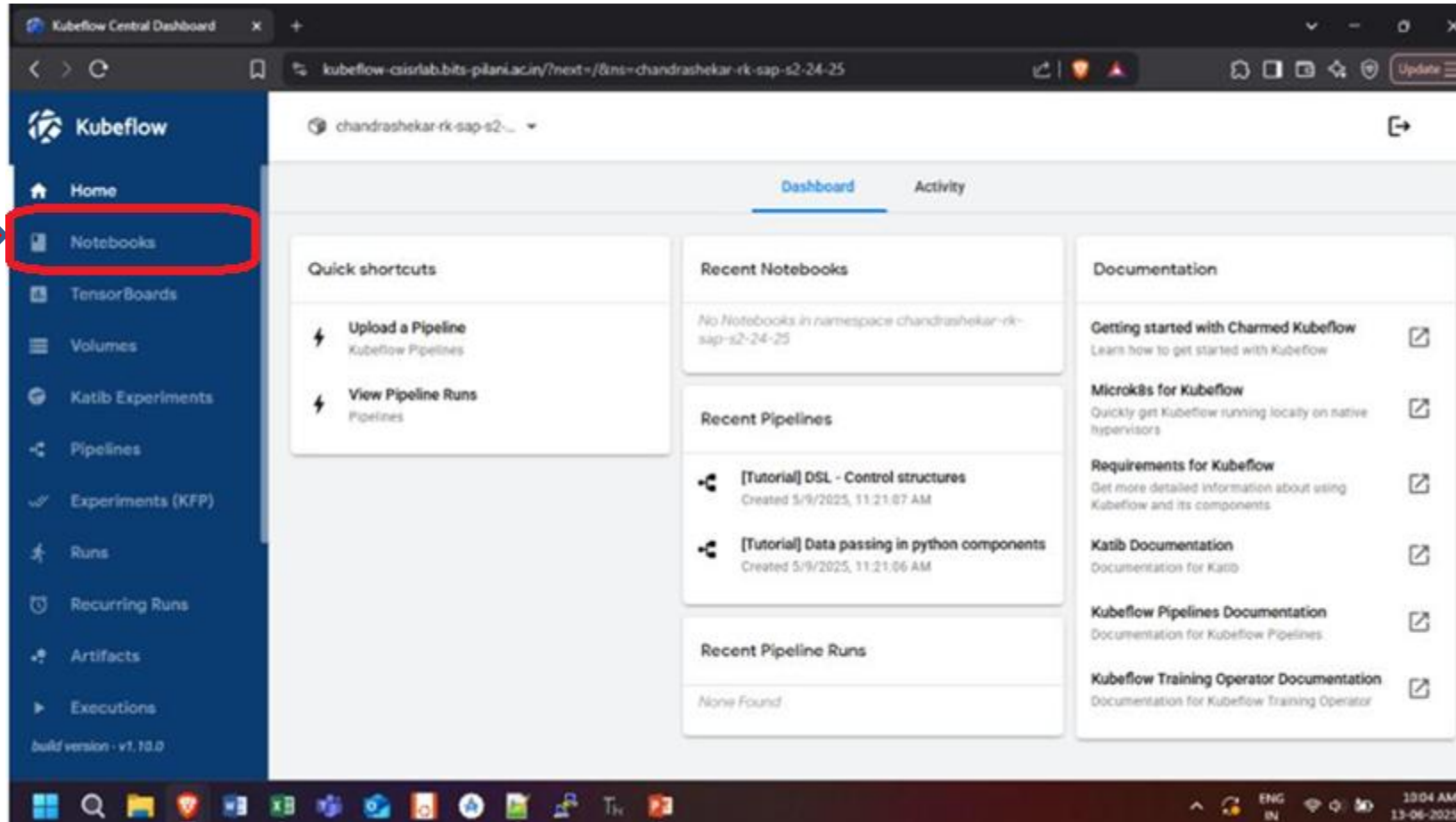
The screenshot shows a web browser window with the address bar displaying "idp.bits-pilani.ac.in/idp/Authn/UserPassword". The page features the BITS Pilani logo on the left and the eLearn logo on the right. The main content area contains a login form with the following elements:

- BITS Mail address:** A text input field containing "chandrashekar.rk@pilani.bits-pilani.ac.in". Below the field, a note reads: "eg. 201218ts450@wlp.bits-pilani.ac.in (for students) / john@goa.bits-pilani.ac.in (for faculty/staff)".
- Password:** A password input field with masked characters "*****".
- [Click here for login Instructions](#)
- [Forgot Password?](#)
- Login** button

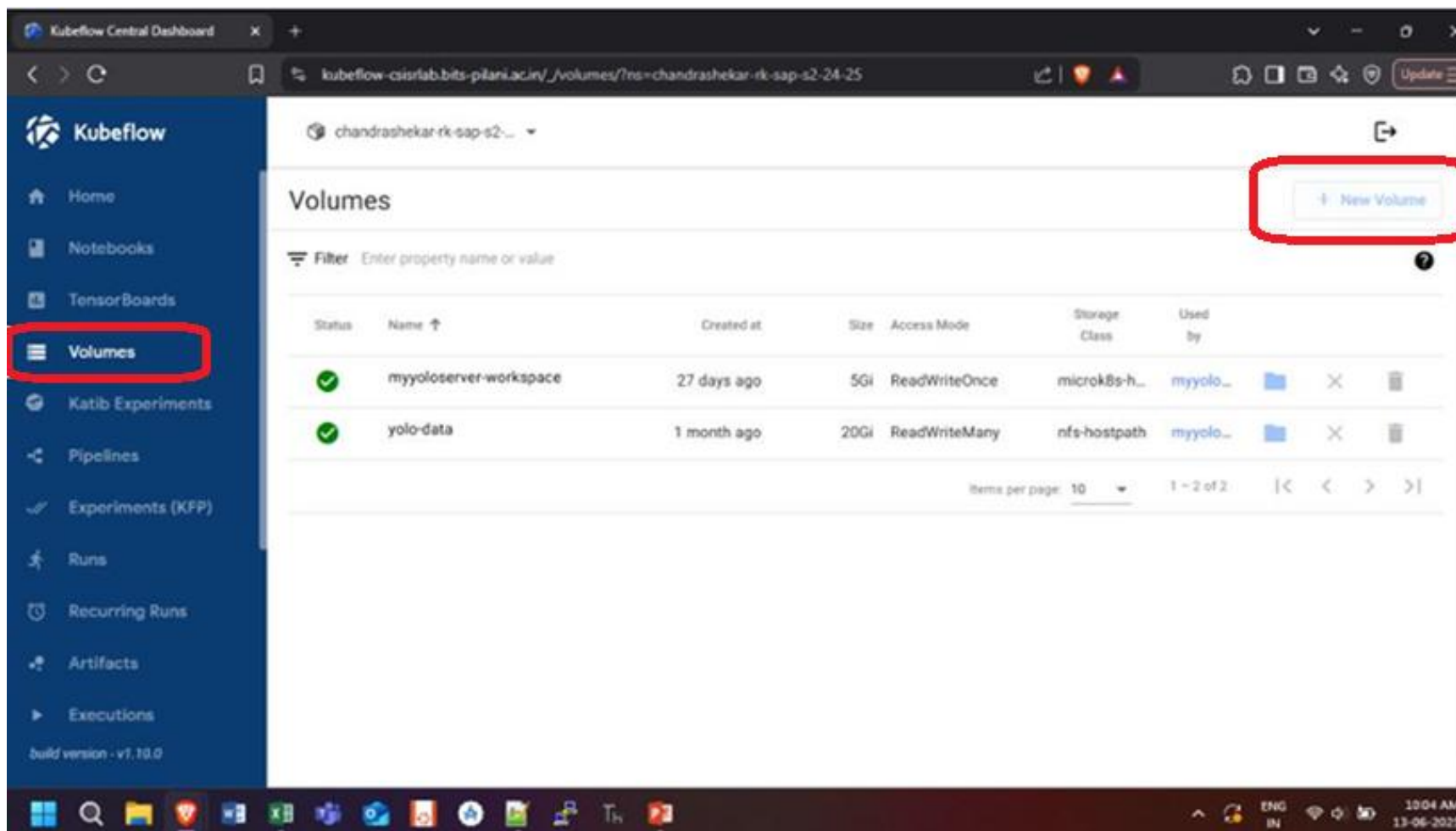
At the bottom of the page, there is a footer with the text: "An institution deemed to be a University estd. vide Sec.3 of the UGC Act,1956 under notification # F.12-23/63.U-2 of June 18,1964" and "Designed and developed by Code Argo". On the right side of the footer, there is a graphic with three colored squares (orange, blue, red) and the words "innovate", "achieve", and "lead" respectively.

The Windows taskbar at the bottom shows various application icons, including the Start menu, Search, File Explorer, and several office applications. The system tray on the right indicates the language is "ENG IN", shows network and volume icons, and displays the time as "10:04 AM" on "13-06-2025".

The Kubeflow portal landing page allows you to create notebooks for running your experiments.



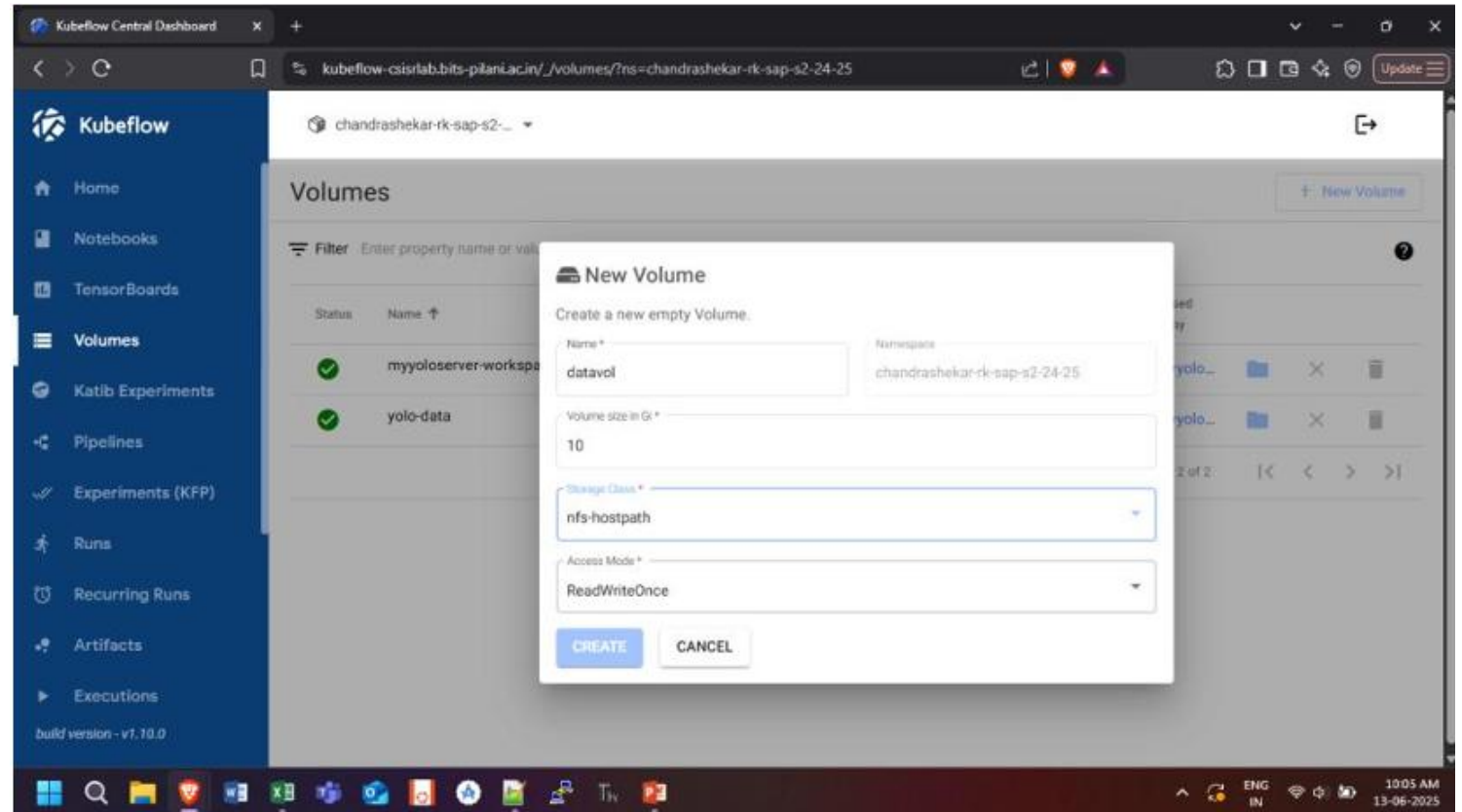
Step 2a. Create Data volume to store your data and source code in a NFS server.



Creating a New Data Volume

- Click on "+ New Volume" (if visible)
- **Name** your volume
- **Volume Size:** Keep the default 10 GB unless more is required (max 30 GB)
- **Storage Class:** Select **nfs-hostpath**
- **Access Mode:** Leave as default ReadWriteOnce
- Click "Create"

Note: Volume will get created only after being attached to a Notebook server



Kubeflow Central Dashboard

kubeflow-csisrlab.bits-pilani.ac.in/_volumes/?ns=chandrashekar-rk-sap-s2-24-25

chandrashekar-rk-sap-s2-...

Volumes

[New Volume](#)

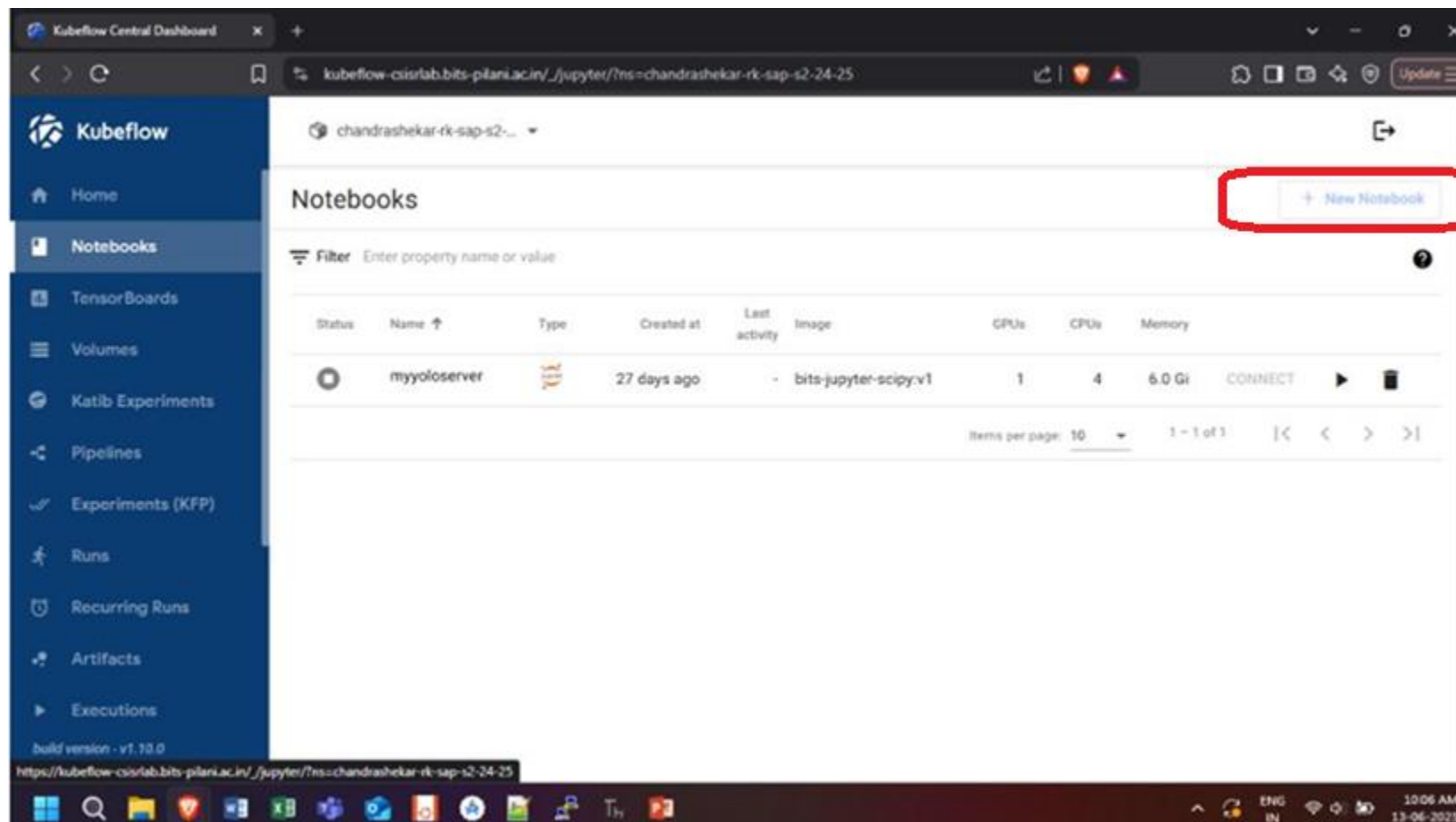
Filter Enter property name or value

Status	Name ↑	Created at	Size	Access Mode	Storage Class	Used by
	datavol	less than a minute ago	10Gi	ReadWriteOnce	nfs-hostpath	
Pending: This volume will be bound when its first consumer is created. E.g., when you first browse its contents, or attach it to a notebook server						
	yolo-data	1 month ago	20Gi	ReadWriteMany	nfs-hostpath	myyolo...

Items per page: 10 1 - 3 of 3

Step 2b: Create Notebook server

Click on “+ New Notebook”



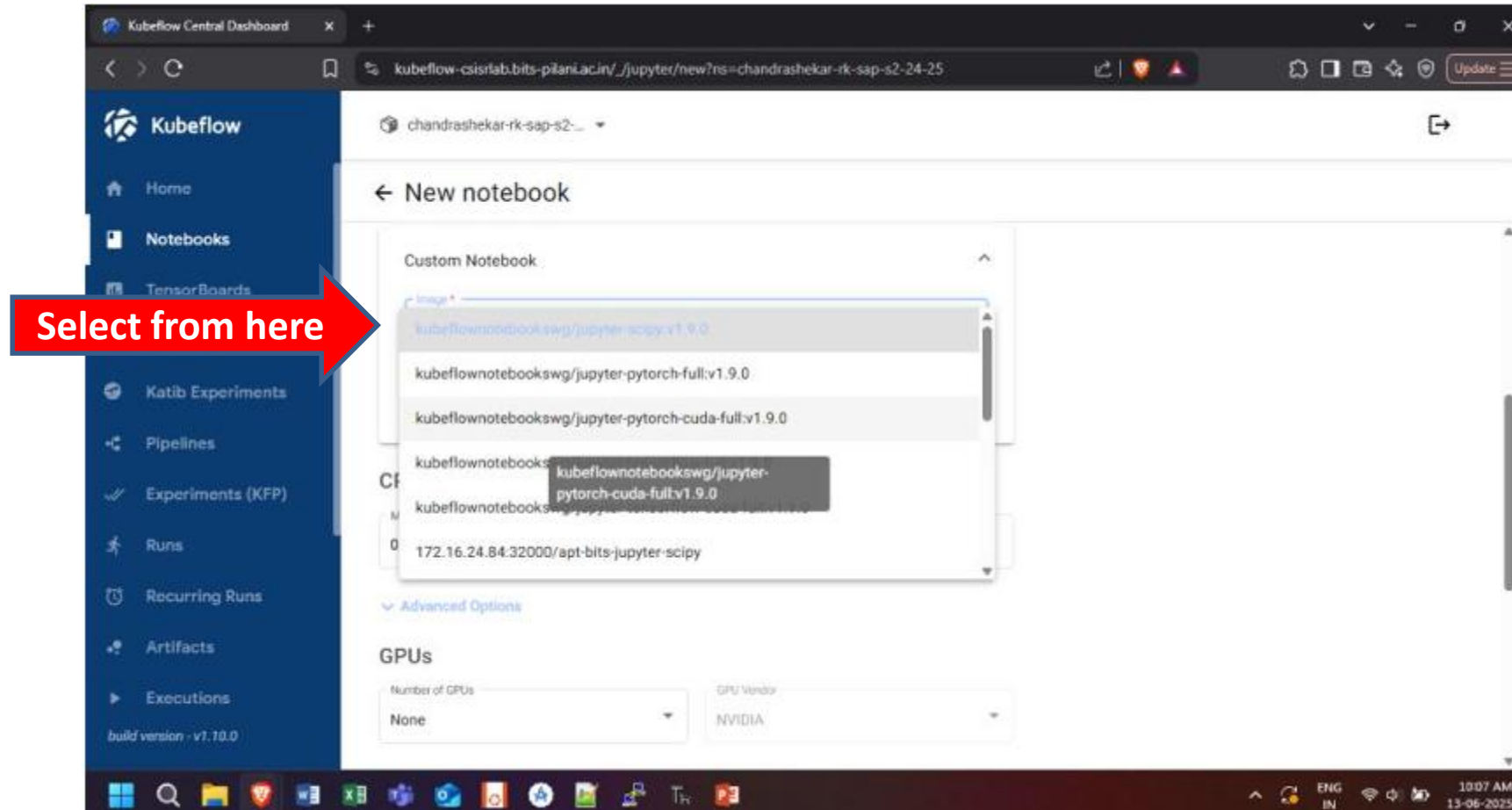
Enter a “Name” and Select “JupyterLab”

The screenshot shows the Kubeflow Central Dashboard interface. On the left is a dark blue sidebar with navigation links: Home, Notebooks, Volumes, Katib Experiments, Pipelines, Runs, Recurring Runs, Artifacts, and Executions. The main content area is titled 'New notebook' and features a 'Name' input field with the text 'demo'. Below this is a grid of three options: JupyterLab (highlighted with a red arrow), VisualStudio Code (labeled '1'), and RStudio (labeled '2'). A 'Custom Notebook' option is visible at the bottom of the grid. The bottom of the dashboard shows 'CPU / RAM' information. The browser's address bar shows the URL 'kubeflow-csislab.bits-pilani.ac.in/_jupyter/new?ns=chandrashekar-rk-sap-s2-24-25'. The Windows taskbar at the bottom indicates the time is 10:06 AM on 13-06-2025.

Enter the name

Select JupyterLab

Select your preferred “Custom Notebook” from the dropdown.



CPU/RAM

- Enter Minimum(0.5)/Maximum(4) CPU cores. Enter same value for both Min and Max
- Enter Minimum (1)/ Maximum(16) Memory(RAM) in GB. Enter same values for both Min and Max

The screenshot shows the 'New notebook' page in the Kubeflow Central Dashboard. The left sidebar contains navigation links: Home, Notebooks, TensorBoards, Volumes, Katib Experiments, Pipelines, Experiments (KFP), Recurring Runs, and Executions. The main content area is titled 'New notebook' and includes a 'Custom Notebook' section with an 'Image' dropdown set to '172.16.24.84:32000/apt-bits-jupyter-scipy'. Below this is the 'CPU / RAM' section, which has four input fields: 'Minimum CPU *' (value: 2), 'Minimum Memory Gi *' (value: 4), 'Maximum CPU' (value: 2), and 'Maximum Memory Gi' (value: 4). Two red arrows with white text point to these fields: 'Enter CPU' points to the 'Minimum CPU' field, and 'Enter Memory' points to the 'Maximum Memory Gi' field. Below the CPU/RAM section is an 'Advanced Options' link and a 'GPUs' section. The bottom of the image shows a Windows taskbar with various application icons and a system tray showing 'ENG IN' and the date '13-06-2025'.

Kubeflow Central Dashboard

kubeflow-csisrlab.bits-pilani.ac.in/_jupyter/new?ns=chandrashekar-rk-sap-s2-24-25

chandrashekar-rk-sap-s2-...

← New notebook

Custom Notebook

Image *

172.16.24.84:32000/apt-bits-jupyter-scipy

Advanced Options

CPU / RAM ?

Minimum CPU *	Minimum Memory Gi *
2	4
Maximum CPU	Maximum Memory Gi
2	4

Advanced Options

GPUs

Number of GPUs

GPU Memory

Enter CPU

Enter Memory

build version - v1.10.0

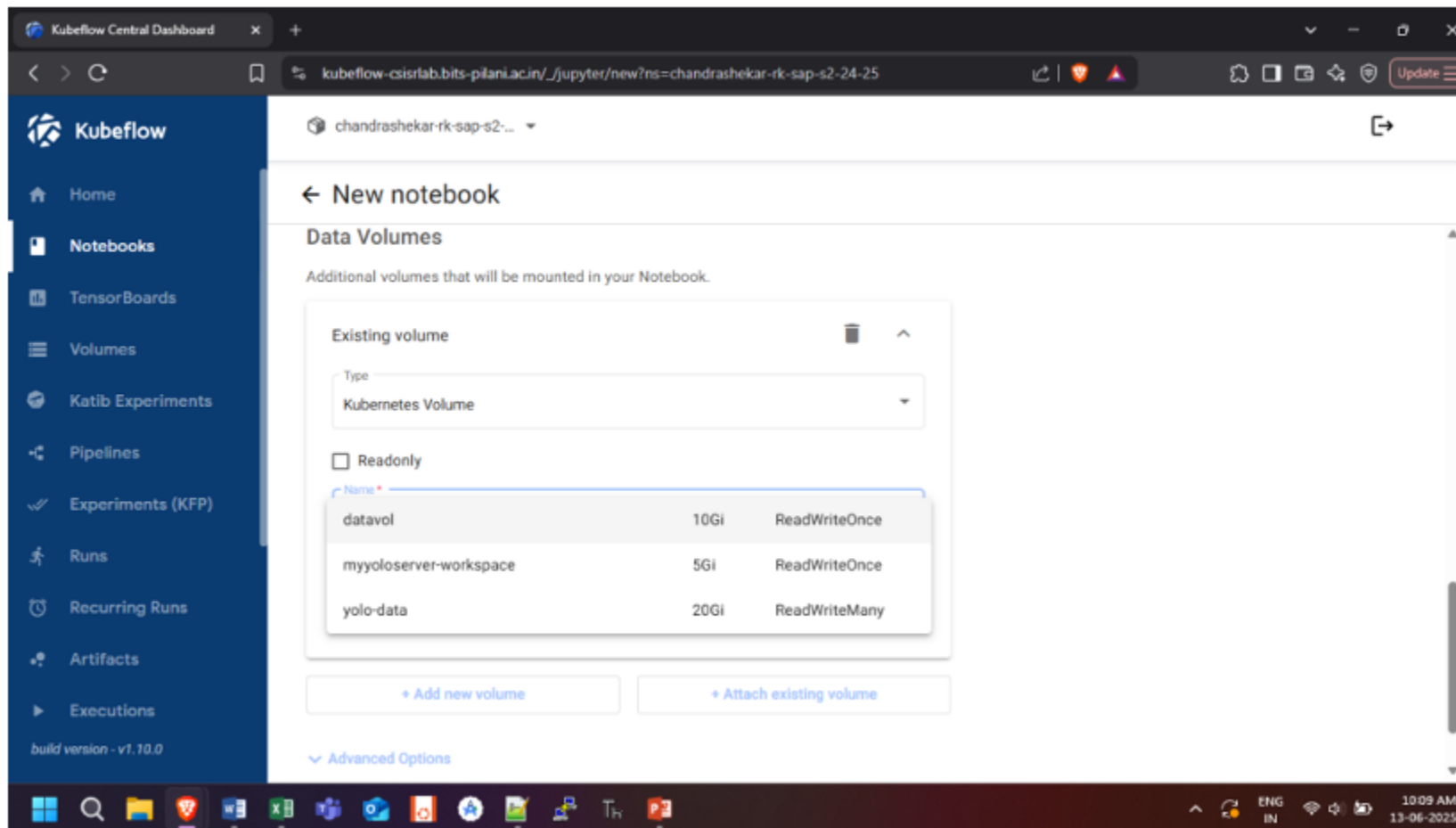
ENG IN 13-06-2025 10:08 AM

Select GPU – 1 (Max allowed)

The screenshot shows the Kubeflow Central Dashboard interface. On the left is a dark blue sidebar with navigation links: Home, Notebooks, TensorBoards, Volumes, Pipelines, Experiments (KFP), Runs, Recurring Runs, Artifacts, and Executions. The main content area is titled 'New notebook' and includes sections for 'GPUs' and 'Data Volumes'. In the 'GPUs' section, the 'Number of GPUs' dropdown menu is open, showing options: None, 1, 2, 4, and 8. A red arrow points from the sidebar to this dropdown with the text 'Select GPU'. To the right, the 'GPU Vendor' dropdown is set to 'NVIDIA', with another red arrow pointing to it from the right with the text 'Select NVIDIA'. Below these are fields for 'Storage' and 'Space, Empty, 5Gi'. The 'Data Volumes' section has buttons for '+ Add new volume' and '+ Attach existing volume'. The bottom of the image shows a Windows taskbar with various application icons and a system clock indicating 10:08 AM on 13-06-2025.

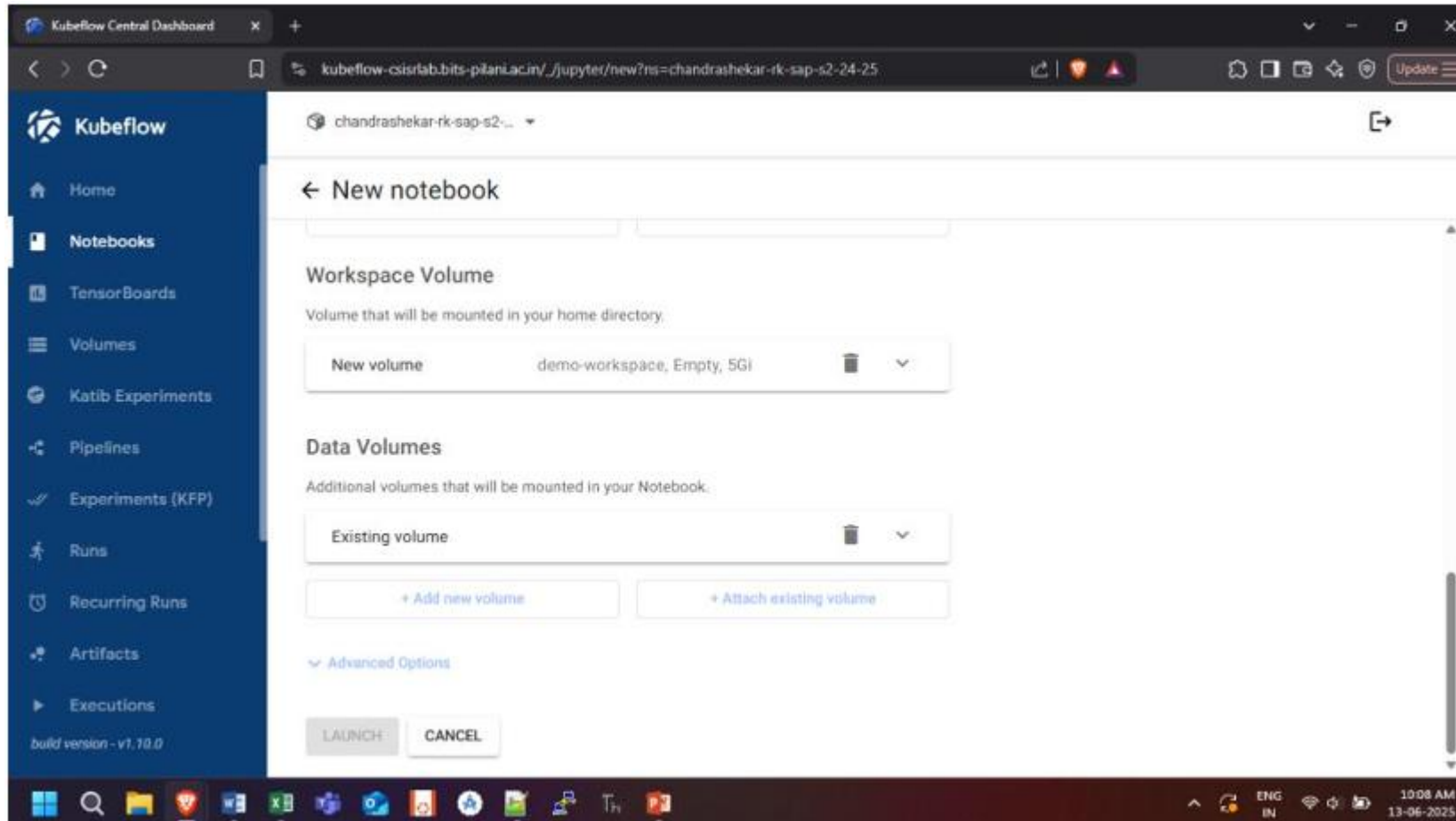
Workspace Volume

Retain the default name and storage space 5GB (if you are installing large packages for training purpose, you may increase the storage space)

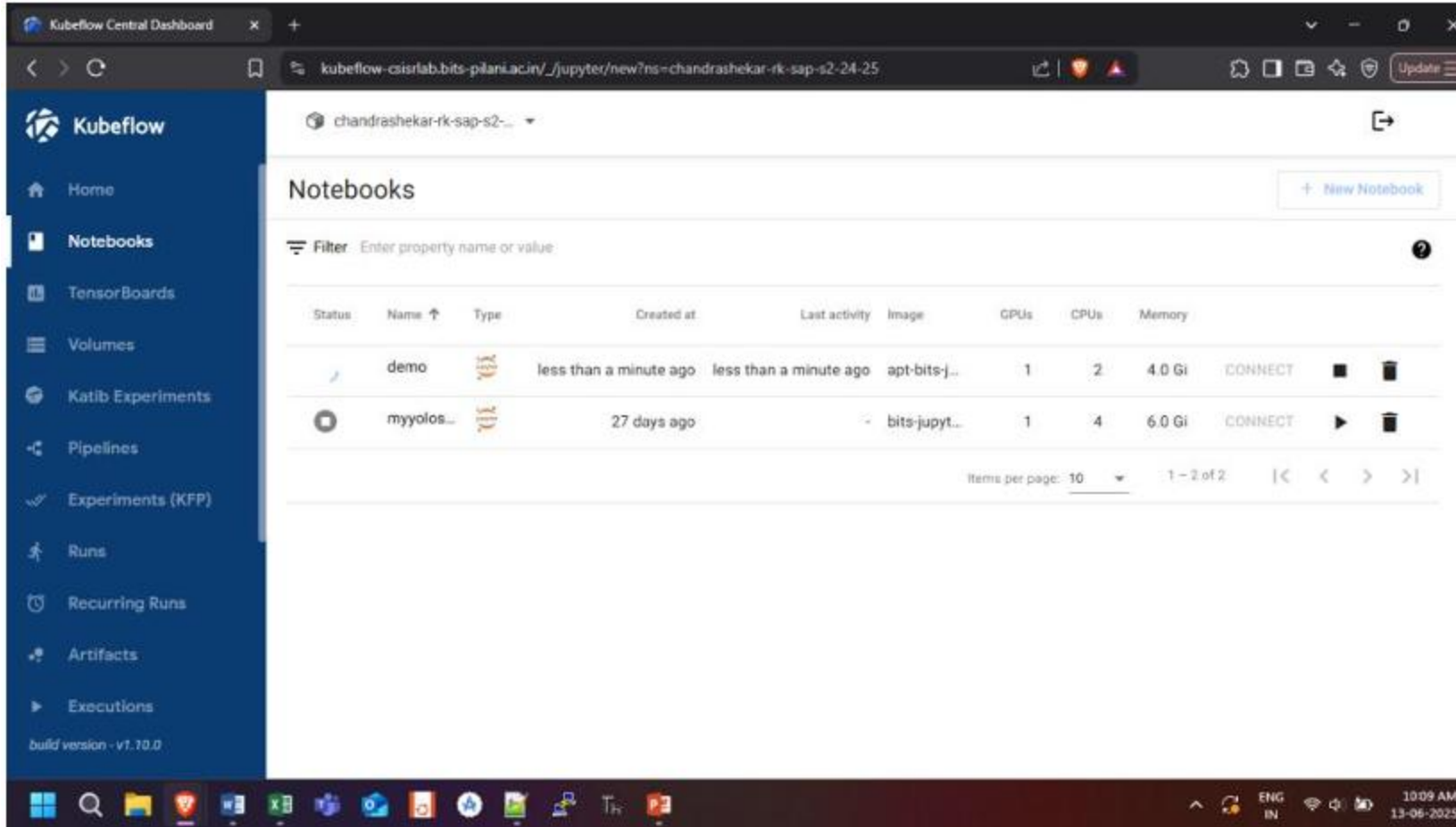


Data Volume:

Click on the dropdown for “Existing Volumes” and select the Data Volume which you have created in Step 2a. Click on Launch button.



Go back to Notebooks to see your Notebook server getting launched.



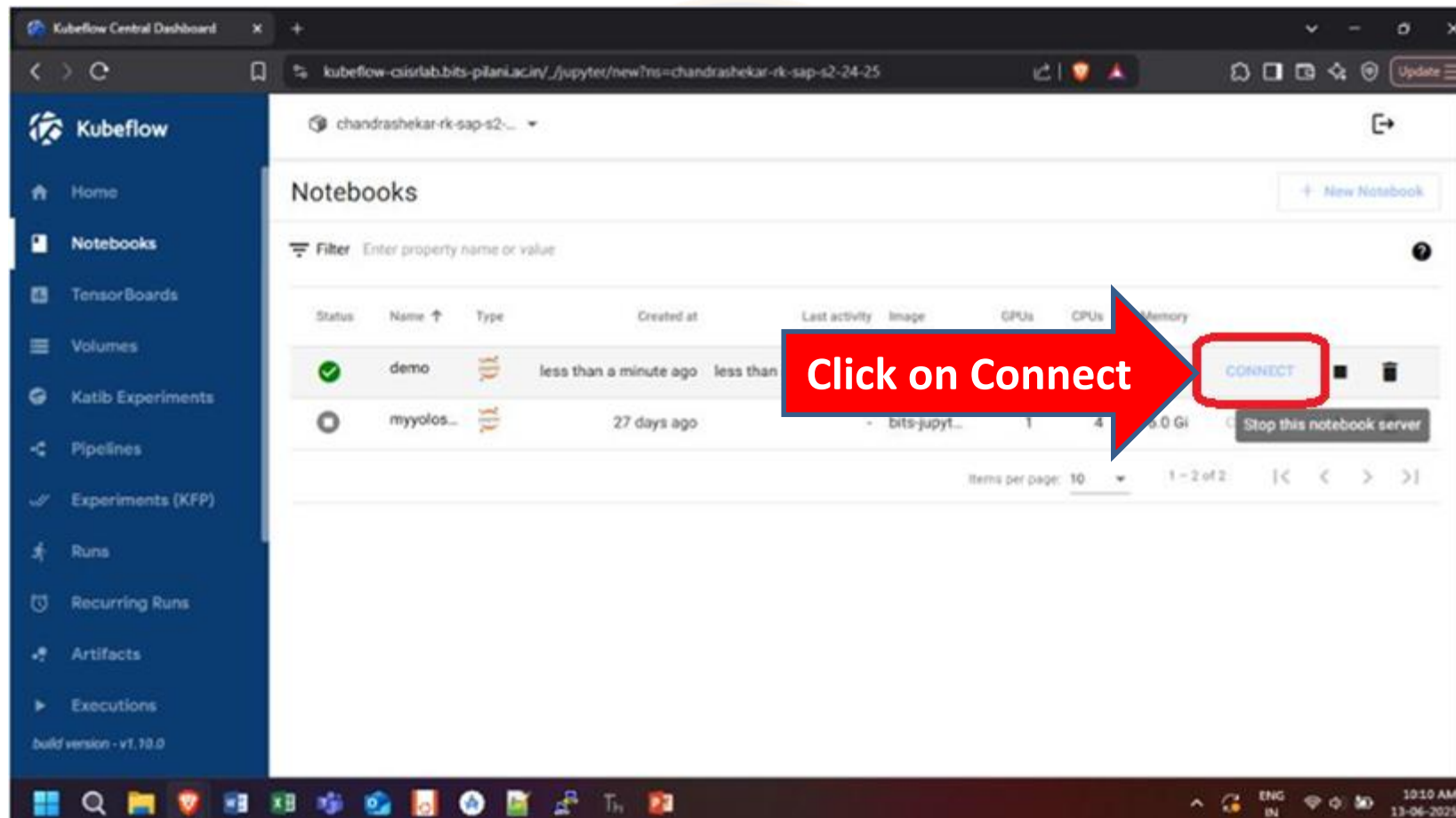
The screenshot shows the Kubeflow Central Dashboard interface. The left sidebar contains navigation links: Home, Notebooks, TensorBoards, Volumes, Katib Experiments, Pipelines, Experiments (KFP), Runs, Recurring Runs, Artifacts, and Executions. The main content area is titled "Notebooks" and includes a "Filter" input field and a "New Notebook" button. Below this is a table listing notebooks:

Status	Name	Type	Created at	Last activity	Image	GPUs	CPUs	Memory	Actions
	demo		less than a minute ago	less than a minute ago	apt-bits-j...	1	2	4.0 Gi	CONNECT
	myyolos...		27 days ago		bits-jupyt...	1	4	6.0 Gi	CONNECT

At the bottom of the table, there is a pagination control showing "Items per page: 10" and "1 - 2 of 2". The Windows taskbar at the bottom indicates the time is 10:09 AM on 13-06-2025.

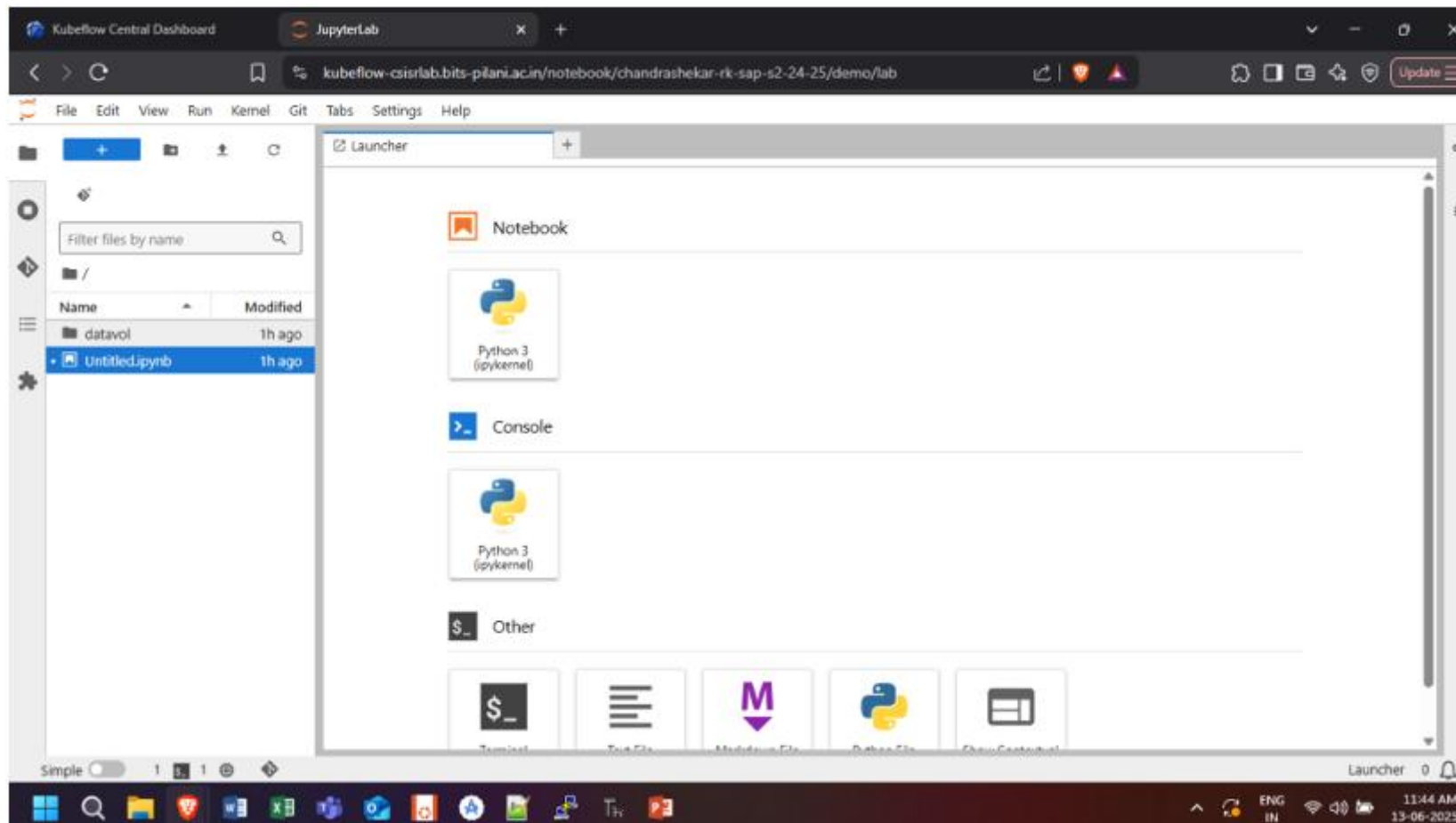
Click on “Connect” button to connect to the Notebook server. Once you have finished your work, click on “STOP” icon.

If you need to delete your Notebook server, click on the “Delete” icon

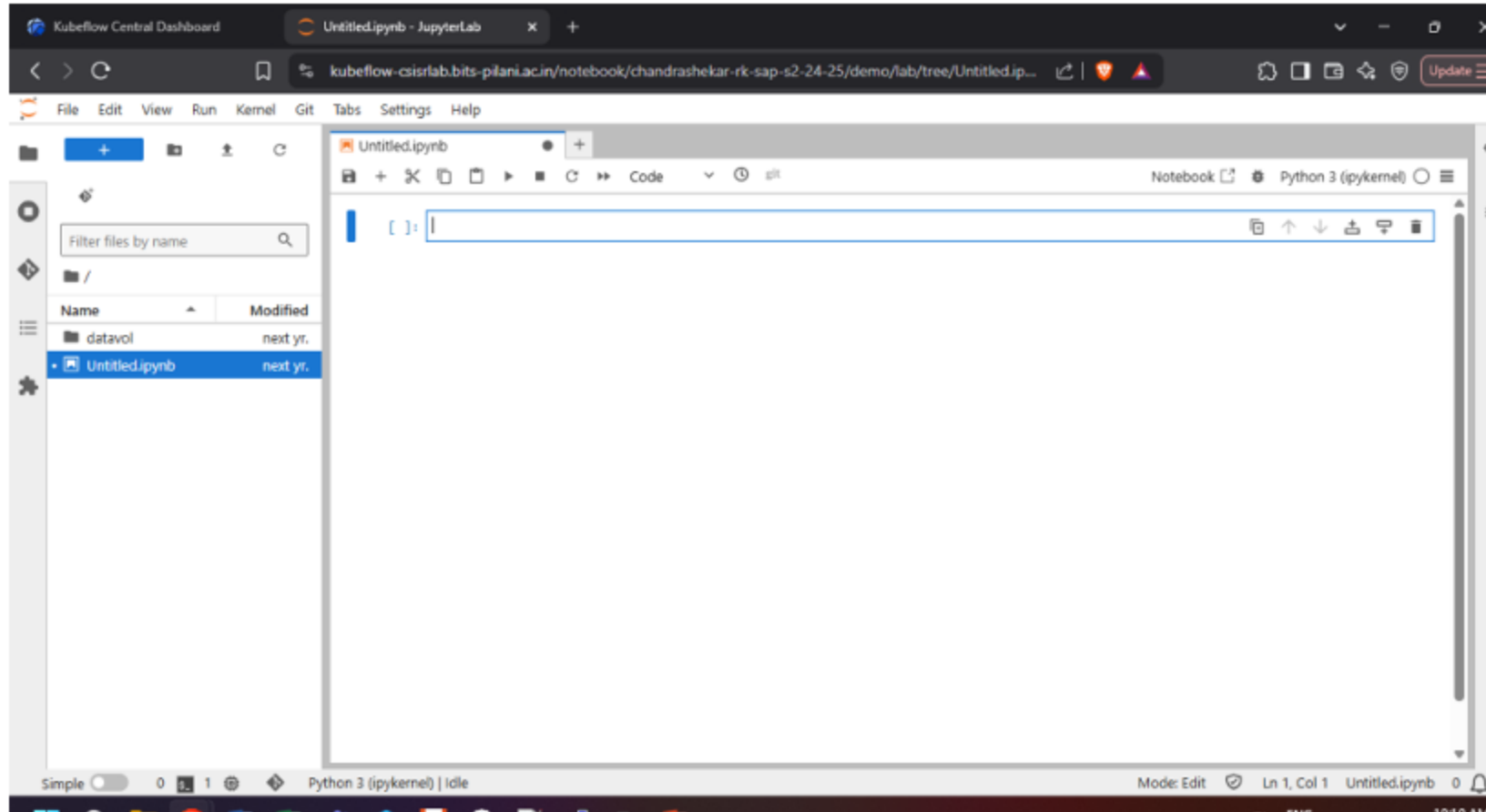


Notebook server interface which can launch – Notebook, Terminal etc.

Left-hand side, you will see the Data Volume attached as a folder to this Notebook server



This is the Notebook interface



You are now ready to use the GPU cluster efficiently. Always manage resources wisely and backup your work!

Notebook interface[Contd..]

You may run Ubuntu(Linux) commands here.

- Create virtual environment
- Install packages using pip install
- Zip your files or folders to download to your local machine

Recursively Zip a Directory into one archive

```
zip -r [file_name.zip] [directory_name]
```

- Unzip
unzip [file_name.zip]



Technical Support details

Technical SPOC: **Mr. Bangaru Santosh Kumar**
Mail id: **csislabsupport@wilp.bits-pilani.ac.in**

Thank You